



Vishal S R 250701844-2025-CSE ▾

V2**Started on** Friday, 21 August 2026, 12:09 AM**State** Finished**Completed on** Saturday, 5 September 2026, 1:34 AM**Time taken** 15 days 1 hour**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeros(int arr[], int low, int high) {
4
5
6      if(low == high) {
7          if(arr[low] == 0)
8              return 1;
9          else
10             return 0;
11     }
12
13
14     int mid = (low + high) / 2;
15
16
17     int left = countZeros(arr, low, mid);
18     int right = countZeros(arr, mid + 1, high);
19
20
21     return left + right;
22 }
23
24 int main() {
25     int n;
26     scanf("%d", &n);
27
28     int arr[n];
29
30     for(int i = 0; i < n; i++) {
31         scanf("%d", &arr[i]);
32     }
33
34     int count = countZeros(arr, 0, n - 1);
35
36     printf("%d", count);
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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